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CONNECTOR

Abstract

A connector includes: a first terminal; a second terminal electrically connected to the first terminal; a screw member configured to electrically connect the first terminal and the second terminal by fastening and provided integrally with one terminal of the first terminal and the second terminal; and a nut rotatably and integrally assembled with the other terminal of the first terminal and the second terminal and fastened to the screw member.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
[0001] The present application is based on and claims the benefit of priority from the prior Japanese Patent Application No. 2024-026310, filed on Feb. 26, 2024, the entire contents of which are incorporated by reference herein.
TECHNICAL FIELD
[0002] The disclosure relates to a connector.
BACKGROUND
[0003] JP 2020-78971 A describes a connector including a bus bar as a first terminal, a motor terminal as a second terminal electrically connected to the bus bar, and a screw member for electrically connecting the bus bar and the motor terminal by fastening. In this connector, the screw member is fastened in a state where the bus bar and the motor terminal are overlapped with each other. By fastening the screw member, the bus bar and the motor terminal are brought into contact with each other, and the bus bar and the motor terminal are electrically connected to each other.
SUMMARY
[0004] In the above connector, the screw member which is a separate component is fastened to the first terminal and the second terminal in a state where the first terminal and the second terminal are overlapped with each other. Therefore, it is necessary to manage and handle the screw member separately from managing and handling the first terminal and the second terminal, and the assemblability is low.
[0005] The disclosure is directed to a connector with improved assemblability.
[0006] A connector in accordance with some embodiments includes: a first terminal; a second terminal electrically connected to the first terminal; a screw member configured to electrically connect the first terminal and the second terminal by fastening and provided integrally with one terminal of the first terminal and the second terminal; and a nut rotatably and integrally assembled with the other terminal of the first terminal and the second terminal and fastened to the screw member.
[0007] According to the above configuration, the assemblability of the connector can be improved.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0008] FIG. **1** is a perspective view of a connector according to an embodiment.
[0009] FIG. **2** is an exploded perspective view of the connector according to the embodiment.
[0010] FIG. **3** is a cross-sectional view of the connector according to the embodiment.
[0011] FIG. **4** is a cross-sectional view of the connector according to the embodiment when a second terminal is arranged in an inclined manner.
[0012] FIG. **5** is an enlarged view of a main part of FIG. **4**.
[0013] FIG. **6** is a perspective view of a first housing of the connector according to the embodiment.
[0014] FIG. **7** is an exploded perspective view of the first housing of the connector according to the embodiment.
[0015] FIG. **8** is a perspective view of a second housing of the connector according to the embodiment.
[0016] FIG. **9** is an exploded perspective view of the second housing of the connector according to the embodiment.
[0017] FIG. **10** is a perspective view of a cover of the connector according to the embodiment.

[0018] FIG. **11** is an exploded perspective view of the cover of the connector according to the embodiment.

DETAILED DESCRIPTION

[0019] Hereinafter, a connector **1** according to an embodiment will be described in detail with reference to the drawings. Note that dimensional ratios in the drawings are exaggerated for convenience of description, and may be different from the actual ratios.

[0020] As illustrated in FIG. **1**, the connector **1** according to the present embodiment is applied to, for example, an electric component **101** mounted on a vehicle. The connector **1** electrically connects, for example, the electric component **101** and other electric components (not illustrated) mounted on the vehicle, such as a power supply and a device. As illustrated in FIGS. **1** to **3**, in the electric component **101**, for example, a mating terminal (not illustrated) is accommodated inside a casing **103**. The casing **103** is provided with an arrangement portion **105** including a through hole which is formed through a wall where the mating terminal is arranged. The connector **1** is assembled to the arrangement portion **105**, and the electric component **101** and the connector **1** are electrically connected to each other.

[0021] As illustrated in FIGS. **1** to **11**, the connector **1** includes a housing **3**, a cover **5**, a first terminal **7**, and a second terminal **9**.

[0022] As illustrated in FIGS. **1** to **4**, the housing **3** includes a first housing **11** and a second housing **13**.

[0023] As illustrated in FIGS. **1** to **4**, **6**, and **7**, the first housing **11** is formed of, for example, an insulating material such as a synthetic resin. The first housing **11** is formed in a tubular shape partially insertable into the arrangement portion **105** of the electric component **101**. On the outer circumference of the first housing **11**, a plurality of (two in the present embodiment) component fixers **15** protruding from the arrangement portion **105** of the electric component **101** to the outer diameter side are provided. A collar **19** into which a bolt **17** to be fastened to a screw hole **107** provided in the casing **103** of the electric component **101** is inserted is insert-molded in the component fixer **15**. On a surface of the component fixer **15** facing the casing **103** of the electric component **101**, a positioning pin **21** to be inserted into a positioning hole **109** provided in the casing **103** protrudes toward the casing **103**. The component fixer **15** positions the first housing **11** with respect to the electric component **101** in a state where a part of the first housing **11** is inserted into the arrangement portion **105** of the electric component **101** by inserting the positioning pin **21** into the positioning hole **109**. The component fixer **15** fixes the first housing **11** to the electric component **101** by fastening the bolt **17** to the screw hole **107** via the collar **19**.

[0024] An inner housing **23** is accommodated in the first housing **11** from the electric component **101** side. The inner housing **23** is formed of, for example, an insulating material such as a synthetic resin. The inner housing **23** is held inside the first housing **11** via engaging portions (not illustrated) that engage with each other. The inner housing **23** accommodates, for example, a plurality of (two in the present embodiment) signal terminals **27** electrically connected to terminal portions of a plurality of (two in the present embodiment) signal lines **25** electrically connected to the electric component **101**. A terminal holder **29** that holds the signal terminal **27** therein is assembled to the inner housing **23**. Note that a locking lance that engages with the signal terminal **27** and holds the signal terminal **27** may be provided inside the inner housing **23**. A packing **31** is assembled in close contact with the outer circumference of the first housing **11** on the electric component **101** side. The packing **31** is held on the first housing **11** by a front holder **33** assembled to the first housing **11** via engagement portions (not illustrated) that engage with each other. The packing **31** is in close contact with the casing **103** of the electric component **101** in a state where the first housing **11** is assembled to the electric component **101**, and seals between the first housing **11** and the electric component **101**. A plurality of (two in the present embodiment) the first terminals **7** are accommodated in the first housing **11** from the side opposite to the electric component **101**.

[0025] As illustrated in FIGS. **1** to **4**, **8**, and **9**, the second housing **13** is formed of, for example, an

insulating material such as a synthetic resin. The second housing **13** has an upper portion formed in a tubular shape into which a part of the first housing **11** can be fitted, and a lower portion formed in a tubular shape communicating with the upper portion and extending in a direction orthogonal to the direction of fitting between the first housing **11** and the second housing **13**. A plurality of (two in the present embodiment) the second terminals **9** are inserted into the lower portion of the second housing **13** from below, and the plurality of second terminals **9** are accommodated in the upper portion of the second housing **13**. On the outer circumference of the upper portion of the second housing **13**, a plurality of (two in the present embodiment) component fixers **35** protruding from the arrangement portion **105** of the electric component **101** to the outer diameter side are provided. A collar **39** into which a bolt **37** to be fastened to a screw hole **111** provided in the casing **103** of the electric component **101** is inserted is insert-molded in the component fixer **35**. The component fixer **35** fixes the second housing **13** to the electric component **101** by fastening the bolt **37** to the screw hole **111** via the collar **39** in a state where the first housing **11** and the second housing **13** are fitted to each other. Note that the first housing **11** and the second housing **13** may be provided with engagement portions that engage with each other to hold the fitted state between the first housing **11** and the second housing **13**.
[0026] A front packing **41** is assembled in close contact with the upper portion of the second housing **13** on the first housing **11** side. The front packing **41** is held on the second housing **13** by a front holder **43** assembled to the second housing **13** via engagement portions (not illustrated) that engage with each other. In the front holder **43**, a plurality of (two in the present embodiment) holding pins **45** protrude toward the second terminal **9** accommodated in the second housing **13**. The front packing **41** is in close contact with the first housing **11** in a state where the first housing **11** and the second housing **13** are fitted to each other, and seals between the first housing **11** and the second housing **13**. A rear packing **47** is assembled in close contact with the upper portion of the second housing **13** on the cover **5** side. The rear packing **47** is held on the second housing **13** by a rear holder **49** assembled to the second housing **13** via engagement portions (not illustrated) that engage with each other. The rear packing **47** is in close contact with the cover **5** in a state where the second housing **13** is covered with the cover **5**, and seals between the second housing **13** and the cover **5**.
[0027] As illustrated in FIGS. **1**, **2**, **10**, and **11**, the cover **5** is formed of, for example, an insulating material such as a synthetic resin. The cover **5** covers the outer circumference of the upper portion of the second housing **13** on the side opposite to the electric component **101**, and is formed in a shape capable of closing the opening of the second housing **13**. On the outer circumference of the cover **5**, a plurality of (two in the present embodiment) component fixers **51** protruding from the arrangement portion **105** of the electric component **101** to the outer diameter side are provided at the same positions as the component fixers **35** of the second housing **13**. A collar **53** into which the bolt **37** to be fastened to the screw hole **111** provided in the casing **103** of the electric component **101** is inserted is insert-molded in the component fixer **51**. The component fixer **51** fixes the cover **5** to the electric component **101** by fastening the bolt **37** to the screw hole **111** via the collar **53** in a state where the cover **5** covers the second housing **13**. By providing the component fixers **51** at the same positions as the component fixers **35** of the second housing **13**, the cover **5** can be fixed by the same bolt **37**, and the number of components can be reduced. Note that the second housing **13** and the cover **5** may be provided with engagement portions that engage with each other to hold a state where the cover **5** covers the second housing **13**. The cover **5** accommodates a short terminal **55** that electrically connects a plurality of (two in the present embodiment) the signal terminals **27** accommodated in the first housing **11**, for example, by press fitting or the like. When the cover **5** is assembled to the electric component **101**, the short terminal **55** electrically connects the plurality of signal terminals **27**, and detects that the cover **5** is normally assembled to the electric component **101**.
[0028] As illustrated in FIGS. **3**, **4**, and **7**, the first terminal **7** is formed of a conductive material,

and is formed, for example, in a columnar shape. The plurality of (two in the present embodiment) the first terminals **7** are accommodated in the first housing **11**. The arrangement position of the first terminals **7** with respect to the first housing **11** is held, for example, by press fitting. Mating connectors **57** and terminal connectors **59** each formed by screw holes are provided on both end sides of the first terminals **7**. For example, a mating screw member (not illustrated) formed of a stud bolt is fastened to the mating connector **57**, and is integrally fixed with the first terminal **7**. The mating terminal of the electric component **101** is arranged on the mating screw member, and a mating nut (not illustrated) is fastened to the mating screw member. By fastening the mating nut to the mating screw member, the mating terminal and the first terminal **7** come into contact with each other, and the mating terminal and the first terminal **7** are electrically connected to each other.
[0029] For example, a screw member **61** formed of a stud bolt is fastened to the terminal connector **59**, and the screw member **61** is integrally fixed to the first terminal **7**. By fixing the screw member **61** to the first terminal **7**, the screw member **61** can be handled as one member with the first terminal **7**, and the number of components can be reduced, whereby management and handling can be simplified. Note that the screw member **61** may be formed into one member continuous with the first terminal **7**, for example, by forming the end portion of the first terminal **7** on the terminal connector **59** side in a columnar shape, and forming the outer circumference of the columnar portion in a screw shape. The second terminal **9** is arranged on the screw member **61**.
[0030] As illustrated in FIGS. **3**, **4**, **5**, and **9**, the second terminal **9** is formed of a conductive material, and is formed, for example, in a plate shape. The plurality of (two in the present embodiment) second terminals **9** are accommodated in the second housing **13**. In the second terminal **9**, a pin hole **63** into which the holding pin **45** of the front holder **43** is inserted is provided through the second terminal **9**. By inserting the holding pin **45** into the pin hole **63**, the arrangement position of the second terminal **9** with respect to the second housing **13** is maintained. The plurality of second terminals **9** are electrically connected to, for example, terminal portions of a plurality of (two in the present embodiment) cables **65** electrically connected to other electric components (not illustrated) by crimping, welding, or the like. A rubber plug **67** is assembled in close contact with the outer circumference of the cable **65** on the second terminal **9** side. The rubber plug **67** is held to the second housing **13** by a lower holder **69** assembled to the second housing **13** via engagement portions (not illustrated) that engage with each other. The lower holder **69** is divided into two and assembled to the cable **65** via engaging portions (not illustrated) that engage with each other. By dividing the lower holder **69**, the assembly position on the cable **65** can be freely determined, and the assemblability can be improved. The rubber plug **67** is in close contact with the second housing **13** in a state where the second terminal **9** is accommodated in the second housing **13**, and seals between the second housing **13** and the cable **65**.
[0031] The second terminal **9** is provided with an arrangement hole **71** formed through the second terminal **9** at a position corresponding to the screw member **61** of the first terminal **7** in a state where the first housing **11** and the second housing **13** are fitted to each other. An inclined surface **73** inclined so as to increase in diameter toward the first terminal **7** side is provided inside the arrangement hole **71**. A nut **75** is arranged in the arrangement hole **71**, and the nut **75** is integrally assembled with the second terminal **9** by, for example, caulking or the like.
[0032] The nut **75** is provided with an insertion portion **77** arranged inside the arrangement hole **71**. An outer diameter of the insertion portion **77** is set to be smaller than an inner diameter of the small-diameter portion of the arrangement hole **71**, and the nut **75** is rotatable with respect to the second terminal **9**. The nut **75** can be inclined with respect to the second terminal **9** by setting the outer diameter of the insertion portion **77** to be smaller than the inner diameter of the small-diameter portion of the arrangement hole **71**. A retaining portion **79** having a larger diameter than the small-diameter portion of the arrangement hole **71** is provided at an end portion of the insertion portion **77**. An inclined surface **81** inclined so as to increase in diameter toward the second terminal **9** side is provided on the retaining portion **79**. When the nut **75** is about to be detached from the

second terminal **9**, the inclined surface **81** abuts on the inclined surface **73** of the arrangement hole **71**, and the retaining portion **79** prevents the nut **75** from being detached from the second terminal **9**. The inclined surface **81** of the retaining portion **79** abuts on the inclined surface **73** of the arrangement hole **71**, thereby restricting the inclination of the nut **75** with respect to the second terminal **9**. By providing the inclined surface **73** in the arrangement hole **71** and providing the inclined surface **81** in the retaining portion **79**, the length of the insertion portion **77** can be shortened, and the nut **75** can be downsized. A restrictor **83** having a diameter larger than the diameter of the arrangement hole **71** is provided in a portion of the nut **75** located outside the arrangement hole **71**. The restrictor **83** is arranged at a position where a gap is formed between the restrictor **83** and an outer surface of the second terminal **9** in a state where the inclined surface **81** of the retaining portion **79** and the inclined surface **73** of the arrangement hole **71** abut each other. By forming a gap between the restrictor **83** and the outer surface of the second terminal **9**, the nut **75** can be inclined with respect to the second terminal **9**. When the nut **75** is about to be detached from the second terminal **9**, the restrictor **83** abuts on the outer surface of the second terminal **9**, thus preventing the nut **75** from being detached from the second terminal **9**. The abutting of the restrictor **83** on the outer surface of the second terminal **9** restricts the inclination of the nut **75** with respect to the second terminal **9**. Note that the retaining portion **79** and the restrictor **83** may be of any sizes as long as they are set to be larger than the diameter of the arrangement hole **71**, and may be a plurality of protrusions or the like discontinuously arranged in the circumferential direction of the arrangement hole **71** without being continuous in the circumferential direction.

[0033] Note that it is also possible that the arrangement hole **71** is not provided with the inclined surface **73**. In this case, the inner diameter of the arrangement hole **71** is formed uniformly over the thickness direction of the second terminal **9**. Regarding such an arrangement hole **71**, the insertion portion **77** may be formed in any size as long as the outer diameter is smaller than the inner diameter of the arrangement hole **71**, and the length is longer than the thickness of the second terminal **9**. The retaining portion **79** may be of any size and located at any position, as long as the retaining portion **79** has a diameter larger than the diameter of the arrangement hole **71** and is provided at the end portion of the insertion portion **77** exposed from the arrangement hole **71**. The restrictor **83** may be arranged in any manner as long as a gap is formed between the restrictor **83** and the outer surface of the second terminal **9** in a state where the retaining portion **79** and the outer surface of the second terminal **9** abut each other. Even with the arrangement hole **71**, the insertion portion **77**, the retaining portion **79**, and the restrictor **83** as described above, the nut **75** can be assembled in a rotatable and inclinable manner with respect to the second terminal **9**.

[0034] The screw member **61** of the first terminal **7** is arranged on the nut **75** in a state where the first housing **11** and the second housing **13** are fitted to each other. At this time, when the screw member **61** and the nut **75** abut each other, and the second terminal **9** is inclined in the direction intersecting the axis of the screw member **61**, the nut **75** is inclined in the direction intersecting the axis of the screw member **61**. Therefore, the axis of the screw member **61** and the axis of the nut **75** can be arranged substantially on the same line. By fastening the nut **75** to the screw member **61** in a state where the nut **75** is arranged on the screw member **61**, the first terminal **7** and the second terminal **9** come into contact with each other, and the first terminal **7** and the second terminal **9** are electrically connected to each other. By rotatably and integrally assembling the nut **75** with the second terminal **9**, the nut **75** can be handled as one member with the second terminal **9**, and the number of components can be reduced, whereby management and handling can be simplified. By rotatably assembling the nut **75** to the second terminal **9**, the nut **75** can be fastened to the screw member **61** even when the screw member **61** is integrally provided with the first terminal **7**. By arranging the nut **75** on the second terminal **9** so as to be able to incline in the direction intersecting the axis of the screw member **61**, the nut **75** can incline when the second terminal **9** is arranged to incline with respect to the first terminal **7**. When the nut **75** is inclined, the nut **75** can be arranged along the axis of the screw member **61**. Therefore, the inclination of the second terminal **9** can be absorbed, and the nut **75** can be stably fastened to the screw member **61**.

[0035] As an example of such assembly of the connector **1**, first, the first housing **11**, the second housing **13**, and the cover **5** are each assembled.

[0036] In the assembling of the first housing **11**, first, the signal terminal **27** provided at the terminal portion of the signal line **25** is accommodated in the inner housing **23**, and the terminal holder **29** is assembled. Next, the inner housing **23** and the first terminal **7** are accommodated in the first housing **11**. Next, the screw member **61** is fastened to the first terminal **7**. The first terminal **7** to which the screw member **61** is fastened in advance may be accommodated in the first housing **11**. Then, the packing **31** is mounted on the first housing **11**, and the front holder **33** is assembled, thus completing the assembly of the first housing **11**.

[0037] In the assembling of the second housing **13**, first, the nut **75** is assembled to the second terminal **9**, and the second terminal **9** is assembled to the end portion of the cable **65** on which the rubber plug **67** is mounted. Next, the second terminal **9** is accommodated in the second housing **13**, and the lower holder **69** is assembled. Then, the front packing **41** and the rear packing **47** are mounted on the second housing **13**, and the front holder **43** and the rear holder **49** are assembled, thus completing the assembly of the second housing **13**.

[0038] In the assembly of the cover **5**, the short terminal **55** is assembled to the cover **5**, and the assembly of the cover **5** is completed.

[0039] In the assembling of the connector **1**, first, the first housing **11** is arranged in the arrangement portion **105** of the electric component **101** so that the positioning pin **21** is inserted into the positioning hole **109**. Next, the bolt **17** is inserted into the collar **19** of the component fixer **15** of the first housing **11**, the bolt **17** is fastened to the screw hole **107**, and the first housing **11** is assembled to the electric component **101**. Next, the second housing **13** is arranged so that the first housing **11** is fitted inside the upper portion of the second housing **13**. Next, the nut **75** is fastened to the screw member **61** to electrically connect the first terminal **7** and the second terminal **9**. At this time, when the first housing **11** is arranged deep inside the second housing **13**, the second terminal **9** may be inclined, but the inclination of the second terminal **9** can be absorbed by the nut **75** being inclined with respect to the second terminal **9**. Next, the cover **5** is assembled to the second housing **13**. Then, the bolt **37** is inserted into the collar **39** of the component fixer **35** of the second housing **13** and the collar **53** of the component fixer **51** of the cover **5**, and the bolt **37** is fastened to the screw hole **111** to complete the assembly of the connector **1**.

[0040] Such a connector **1** includes the first terminal **7**, the second terminal **9** electrically connected to the first terminal **7**, and the screw member **61** that electrically connects the first terminal **7** and the second terminal **9** by fastening. The screw member **61** is provided integrally with one of the first terminal **7** and the second terminal **9** (the first terminal **7** in the present embodiment). The nut **75** fastened to the screw member **61** is rotatably and integrally assembled with the other terminal (the second terminal **9** in the present embodiment) of the first terminal **7** and the second terminal **9**.

[0041] Since the screw member **61** is provided integrally with the first terminal **7**, the screw member **61** can be handled as one member with the first terminal **7**. Therefore, the number of components can be reduced, and management and handling can be simplified. Since the nut **75** is rotatably and integrally assembled with the second terminal **9**, the nut **75** can be handled as one member with the second terminal **9**. Therefore, the number of components can be reduced, and management and handling can be simplified. By rotatably assembling the nut **75** to the second terminal **9**, the nut **75** can be fastened to the screw member **61** even when the screw member **61** is integrally provided with the first terminal **7**.

[0042] Therefore, in such a connector **1**, it is not necessary to manage and handle the screw member **61** and the nut **75** separately from the first terminal **7** and the second terminal **9**, and the assemblability can be improved.

[0043] The nut **75** is arranged so as to be able to incline with respect to the other terminal (the second terminal **9** in the present embodiment) in a direction intersecting the axis of the screw

member **61**.
[0044] Due to the inclination of the nut **75**, the nut **75** can be arranged along the axis of the screw member **61**, even if the second terminal **9** is arranged to be inclined with respect to the first terminal **7**. Therefore, the inclination of the second terminal **9** can be absorbed, and the nut **75** can be stably fastened to the screw member **61**.
[0045] The other terminal (the second terminal **9** in the present embodiment) is provided with the arrangement hole **71** formed through the second terminal **9**. The nut **75** is provided with the insertion portion **77** having the outer diameter set to be smaller than the inner diameter of the arrangement hole **71** and arranged inside the arrangement hole **71**, and the restrictor **83** having the diameter set to be larger than the diameter of the arrangement hole **71** and arranged outside the arrangement hole **71**. A gap is formed between the restrictor **83** and the outer surface of the second terminal **9**.
[0046] The nut **75** can be inclined with respect to the second terminal **9** in a rotatable manner by setting the outer diameter of the insertion portion **77** to be smaller than the inner diameter of the arrangement hole **71**. By forming the gap between the restrictor **83** and the outer surface of the second terminal **9**, the nut **75** can be inclined with respect to the second terminal **9**. When the nut **75** is inclined, the abutting of the restrictor **83** on the outer surface of the second terminal **9** can restrict the inclination of the nut **75** with respect to the second terminal **9**.
[0047] The retaining portion **79** having the diameter set to be larger than the diameter of the arrangement hole **71** is provided at an end portion of the insertion portion **77**.
[0048] Therefore, the insertion portion **77** does not fall out of the arrangement hole **71**, the nut **75** can be prevented from being detached from the second terminal **9**, and the nut **75** can be stably held by the second terminal **9**.
[0049] In the present embodiment, the screw member **61** is provided integrally with the first terminal **7**, and the nut **75** is rotatably and integrally assembled with the second terminal **9**, but the present invention is not limited thereto. For example, the screw member **61** may be provided integrally with the second terminal **9**, and the nut **75** may be rotatably and integrally assembled with the first terminal **7**.
[0050] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. A connector comprising: a first terminal; a second terminal electrically connected to the first terminal; a screw member configured to electrically connect the first terminal and the second terminal by fastening and provided integrally with one terminal of the first terminal and the second terminal; and a nut rotatably and integrally assembled with the other terminal of the first terminal and the second terminal and fastened to the screw member.
2. The connector according to claim 1, wherein the nut is arranged to be able to incline with respect to the other terminal in a direction intersecting an axis of the screw member.
3. The connector according to claim 2, wherein the other terminal includes an arrangement hole penetrating the other terminal, the nut includes an insertion portion having an outer diameter smaller than an inner diameter of the arrangement hole and arranged inside the arrangement hole, and a restrictor larger than a diameter of the arrangement hole and arranged outside the arrangement hole, and a gap is formed between the restrictor and an outer surface of the other

terminal.

4. The connector according to claim 3, wherein the insertion portion includes an end portion having a retaining portion larger than the diameter of the arrangement hole.